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FULL BORE ELECTRIC FLOW CONTROL VALVE SYSTEM

Abstract

A technique facilitates flow control downhole via at least one flow control valve. According to an example, a flow control valve has an internal piston. Additionally, an electrically powered actuator is mounted externally to the flow control valve and connected to the internal piston via a linkage. The electrically powered actuator is responsive to electrical inputs to shift the internal piston to desired flow positions of the flow control valve.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of U.S. patent application Ser. No. 18/360,172 filed Jul. 27, 2023, which claims priority to U.S. patent application Ser. No. 17/253,147, filed Dec. 17, 2020, which is a national stage of PCT/US2019/038438, filed Jun. 21, 2019, which claims the benefit of U.S. Provisional Application No. 62/688,843, filed Jun. 22, 2018, their entireties of which are incorporated by reference herein and should be considered part of this specification.

BACKGROUND

[0002] An oil well may have multiple production zones or intervals. It is of interest for the operator to be able to produce these zones altogether (commingled production) to maximize production and the return on investment made in such well. The different producing zones may have different pressures and may deplete at different rates. To optimize production or even shut off a water producing zone, the operator relies on downhole flow control valves (FCVs) that control the flow of hydrocarbon from each producing interval into the production tubing string. The same applies for an injection well where selective and controlled injection into the different intervals involves controlling the flow of fluid at each interval.

[0003] FCVs are traditionally hydraulically operated from surface by hydraulic control lines running from in the well and fed through the well head and packers. Because the number of penetrators or allowable control lines is limited, this may restrict the number of valves that can be installed in a well. Moreover, such a well often includes chemical injection lines and electrical cable for communication and power of downhole sensors, thus restricting even further the number of hydraulic penetrations left at the well head or packer.

SUMMARY

[0004] In general, a system and methodology are provided for facilitating flow control downhole. According to an embodiment, a flow control valve has an internal piston. Additionally, an electrically powered actuator is mounted externally to the flow control valve and connected to the internal piston via a linkage. The electrically powered actuator responds to electrical inputs to shift the internal piston to desired flow positions of the flow control valve.

[0005] The flow control valve can include a housing, with the internal piston movably disposed within the housing. The actuator can be held in place along an outer surface of the housing with one or more clamps or protectors. An outer surface of the housing can include one or more grooves. The actuator can be disposed in one of the one or more grooves. The outer surface of the housing can have a first groove housing the actuator and a second groove housing electronics and/or sensors.

[0006] The actuator can be an electro-mechanical actuator (EMA) or an electro-hydraulic actuator (EHA).

[0007] A system including the flow control valve and actuator can further include a pump system and a manifold. The pump system includes a motor and a pump. The manifold includes hydraulic circuitry that links the pump system to the actuator. The pump system is configured to pump hydraulic control fluid from a reservoir through the manifold to the actuator. The manifold can include at least one solenoid operated valve (SOV).

[0008] Mechanical intervention for mechanically shifting the flow control valve can be performed while the actuator is connected to the internal piston of the flow control valve. In some configurations, the linkage can be disconnected to enable mechanical intervention for mechanically shifting the flow control valve.

[0009] The flow control valve can be mounted along a well tubing. The flow control valve can have a flow area equivalent to an internal cross-sectional area of the well tubing.

[0010] In some embodiments, a method of operating a flow control valve includes powering up a pump system configured to pump hydraulic control fluid from a reservoir; activating a selected solenoid operated valve (SOV) in a manifold comprising hydraulic circuitry linking the pump system with an electro-hydraulic actuator mounted externally to the flow control valve; flowing hydraulic control fluid from the reservoir, through the manifold, and into a chamber of the actuator such that a piston of the actuator moves in an open or a close direction; and moving a piston of the flow control valve by movement of the piston of the actuator.

[0011] The SOV can be a 3-way, 2-position, normally closed valve. The SOV can be a 2-way, 2-position, normally open valve. The SOV can act as a directional switch.

[0012] The method can further include performing mechanical intervention on the actuator by using a shifting tool to mechanically move the piston of the actuator.

[0013] In some embodiments, a flow control valve includes a housing; a piston movably disposed within the housing to adjust flow through the flow control valve; at least one groove formed in an outer surface of the housing, the at least one groove housing an electrically powered actuator; and a linkage coupling the actuator to the piston such that movement of the actuator causes movement of the piston.

[0014] The at least one groove can include a first groove housing the actuator and a second groove housing electronics. The actuator can be an electro-hydraulic actuator. The electro-hydraulic actuator can include an internal piston. In use, movement of the internal piston of the actuator causes movement of the piston of the flow control valve to adjust flow through the flow control valve.

[0015] However, many modifications are possible without materially departing from the teachings of this disclosure. Accordingly, such modifications are intended to be included within the scope of this disclosure as defined in the claims.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0016] Certain embodiments of the disclosure will hereafter be described with reference to the accompanying drawings, wherein like reference numerals denote like elements. It should be understood, however, that the accompanying figures illustrate the various implementations described herein and are not meant to limit the scope of various technologies described herein, and:

[0017] FIG. 1 is a cross-sectional illustration of an example of a flow control valve having a housing, a piston, a choke, and choke seals, according to an embodiment of the disclosure;

[0018] FIG. 2 is an illustration of a flow control valve architecture with an actuator implanted in a main housing, according to an embodiment of the disclosure;

[0019] FIG. 3 is a cross-sectional view of a flow control valve showing a housing containing actuators, electronics, and sensors, according to an embodiment of the disclosure;

[0020] FIG. 4 is an illustration of an example of a flow control valve with electronics and sensors located in grooves of a main housing, according to an embodiment of the disclosure;

[0021] FIG. 5 is an illustration of an example of an electro-mechanical actuator for use with a flow control valve, according to an embodiment of the disclosure;

[0022] FIG. 6 is an illustration of an in-line translating axle which may be used with the electro-mechanical actuator of FIG. 5, according to an embodiment of the disclosure;

[0023] FIG. 7 is an illustration of an example of an electro-hydraulic actuator for use with a flow control valve, according to an embodiment of the disclosure;

[0024] FIG. 8 is an illustration of another example of an electro-hydraulic actuator for use with a

flow control valve, according to an embodiment of the disclosure;

[0025] FIG. **9** is a schematic illustration of an example of an electro-hydraulic actuator and associated hydraulic circuitry for use with a flow control valve, according to an embodiment of the disclosure;

[0026] FIGS. **10A-10D** are schematic illustrations of examples of the electro-hydraulic actuator and associated hydraulic circuitry as illustrated in FIG. **9** in various operational modes, according to an embodiment of the disclosure;

[0027] FIG. **11** is a schematic illustration of another example of an electro-hydraulic actuator and associated hydraulic circuitry for use with a flow control valve, according to an embodiment of the disclosure;

[0028] FIGS. **12A-12D** are schematic illustrations of examples of the electro-hydraulic actuator and associated hydraulic circuitry as illustrated in FIG. **11** in various operational modes, according to an embodiment of the disclosure;

[0029] FIG. **13** is a schematic illustration of another example of an electro-hydraulic actuator and associated hydraulic circuitry for use with a flow control valve, according to an embodiment of the disclosure; and

[0030] FIGS. **14A-14D** are schematic illustrations of examples of the electro-hydraulic actuator and associated hydraulic circuitry as illustrated in FIG. **13** in various operational modes, according to an embodiment of the disclosure.

DETAILED DESCRIPTION

[0031] In the following description, numerous details are set forth to provide an understanding of some embodiments of the present disclosure. However, it will be understood by those of ordinary skill in the art that the system and/or methodology may be practiced without these details and that numerous variations or modifications from the described embodiments may be possible.

[0032] The disclosure herein generally involves a system and methodology to facilitate flow control downhole. According to embodiments, the system and methodology provide mechanical architectural elements for the design of an electrically powered downhole flow control valve (FCV). A solid gauge mandrel type design for a FCV may restrict the maximum allowable production flow rate through the valve. In contrast, FCVs according to the present disclosure can have a flow area that may be equivalent to the tubing internal cross section.

[0033] Various embodiments described herein cover options for integrating an Electro-Mechanical Actuator (EMA), mounted externally to the valve, and connecting it to the FCV internal piston. This permits use of a traditional FCV choke design with an internal piston, a hard erosion resistant sleeve for the flow openings, and existing choke sealing elements. Embodiments also cover the implementation of an Electro-Hydraulic Actuator (EHA) in lieu of the EMA. As the available power source for the actuator is electrical, the EHA also may include a hydraulic fluid reservoir and an electrically powered pump to provide the pressurized hydraulic fluid. In addition, the present disclosure provides several options for controlling the position of FCV while actuated with the EHA or EMA. Various embodiments described herein relate to the linkage between the actuator and the FCV internal piston, in the case of an EMA drive. The linkage system may include options for a disconnect ability in case it is desired to mechanically intervene and operate the valve through slickline or other mechanical intervention methods.

[0034] In subsea fields, hydraulic flow control valves utilize the infrastructure on the seabed to handle and distribute pressurized hydraulic fluid to each well head and each hydraulic control line. In conventional systems, this functionality represents a substantial cost and complexity for the subsea infrastructure, the umbilical, and the surface platform or FPSO. Removing the need to handle pressurized hydraulic fluid can lead to substantial reduction in cost of the subsea infrastructure.

[0035] A fully electric downhole flow control system helps overcome both of these limitations especially when other (traditionally hydraulically operated) equipment in the well is converted to

full electric as well (e.g. the safety valve). A high number of electrically powered flow control devices can be connected on a single electrical cable, thus using just one penetrator at the wellhead. Electrical power is used to operate such a completion system, simplifying greatly the system on the seabed and potentially also simplifying the umbilical to the production facility.

[0036] A valve providing a flow area equivalent to the tubing inner cross-sectional area is referred to as a “Full Bore” valve. Traditional hydraulic full bore valves have an internal piston to control the amount of opening and flow through a choke. Given the size of the piston, sealing systems and bearings around the piston, substantial loads may be used to operate such a valve by overcoming the amount of friction generated by the dynamic and choke seals. Hydraulically operated valves can easily provide the desired load via a high hydraulic supply pressure and a large piston area. Converting such valves to an electric drive poses some challenges as the load provided by an electromechanical actuator is usually lower than what can be delivered by traditional hydraulic FCVs.

[0037] One way to address this challenge is to implement the electric drive on a smaller valve, such as a side-pocket mandrel valve. In such an arrangement, the choke, piston and scaling systems are much smaller and utilize substantially less force, at the expense of a reduced flow area and limited maximum allowable flow rate through the valve. For applications involving high flow rates, the challenge is to find a suitable way of integrating an electrically powered actuator mechanism able to deliver sufficient force to operate a full bore valve.

[0038] Referring initially to FIG. 1, embodiments described herein cover architectural choices for designing an electrically powered FCV. Designs according to the present disclosure advantageously use the configuration of traditional FCVs including an internal piston, but also maximize the flow area and are operated electrically. Use of the configuration of traditional FCVs allows for minimizing development effort and takes advantage of a robust choke design already developed for hydraulic full bore FCVs.

[0039] Full bore FCVs may rely on an internal piston moving back and forth, e.g. up or down, to open or close hydraulic flow ports which selectively places the annulus and the tubing in fluid communication. While the upper section of the FCV is dedicated to the actuation and position indexing mechanism, the choking (or flow control) and sealing functions of the valve are done at the choke section. As shown in FIG. 1, the choke **100** may include a sleeve **102**, which can be made of or include a hard material for erosion resistance, and an inner piston **104**, which in operation closes and/or opens ports **106** of the sleeve **102**. The piston **104** and sleeve **102** are disposed in a choke housing **108**. The choke also includes a seal stack **110** sealing off the valve when the piston **104** is in the closed position.

[0040] In FCVs according to the present disclosure, a section, for example, an upper section when deployed in a horizontal portion of a well, of the flow control valve may be modified to house an electrical actuator **200**, for example as shown in FIG. 2. As described herein, the actuator **200** can be an electro-mechanical actuator (EMA) or an electro-hydraulic actuator (EHA). In some configurations, the electrical actuator **200** is housed in a groove cut throughout the FCV main housing **118**, for example, along and/or in an outer surface of the FCV main housing **118**. The internal piston **104** of the valve is able to hold the pressure when the valve is closed due to, for example, two sealing elements in the form of the choke seal(s) or seal stack **110** in the choke housing **108** and a dynamic seal **120** at the top of the main housing **118**. Such implementation allows an externally mounted actuator **200** to connect to the valve internal piston **104** via a linkage mechanism **300**, while at the same time being housed and protected by the main housing **118** itself, as illustrated in FIG. 2. The actuator **200** may be maintained in place by additional clamps and/or protectors **128** as illustrated. The electronics controlling the actuator **200** and/or electronics for telemetry with the surface control panel can be placed in parallel in separate groove(s) in the FCV housing **118** to reduce the overall length of the system.

[0041] As further illustrated in FIG. 3, this configuration also advantageously allows multiple

actuators **200** to be assembled onto the FCV. This could be particularly advantageous for electro hydraulic actuator (EHA) solutions, as described below, in which one assembly including a motor, a pump, and a distribution manifold distributes pressurized hydraulic fluid to multiple actuators **200**, thus increasing the actuation load. In some configurations, multiple EMAs can be connected to a single piston **104**.

[0042] As described, FIG. **3** illustrates the integration of various elements, including multiple actuators **200** and various electronics, in the FCV main housing **118**, each in a separate groove. This schematic shows the housing **118** containing two actuators **200**, electronics **230** for controlling one or both of the actuators **200**, and electronics and/or sensors **240** (e.g., for telemetry with the surface and/or position sensing). As shown, the housing **118** can also house one or more sensors **250** (such as position, pressure, temperature, and/or other sensors or gauges) and/or one or more bypass lines **260**. The FCV main housing **118** is able to resist tensile and compressive loads as the piston **104** alone takes the differential pressure across the valve when closed. This enables machining of the housing **118** to host other sensors as well, such as pressure and temperature sensors **250**, as also illustrated in FIG. **4**. The FCV housing **118** can therefore replace a traditional gauge carrier mandrel, reducing the overall length of intelligent completion smart assemblies (including a FCV and one or more sensors or gauges).

[0043] In various embodiments, the electrically powered actuator **200** driving the FCV can be an electro mechanical actuator (EMA), which receives electrical power as input, e.g., from one or more electrical cables **270** as shown in FIG. **4**, and converts the electrical power into a translating movement. The EMA includes, for example, an electric motor **202**, a gear box or reducer **204**, a screw **206** (e.g., a ball screw or roller screw), and one or more bearings **208**, as shown in the example configuration of FIG. **5**. These internal components or elements operate to convert the electrical power to translational movement. These elements may be immersed in a dielectric fluid providing electrical insulation and lubrication. This oil may be pressure compensated with the external environment by a bellows.

[0044] Referring generally to FIG. **5**, an example of an EMA is illustrated as providing two output pins **210** on the side of the actuator **200** that can be connected to the FCV piston **104** by a linkage mechanism **300**. In another embodiment illustrated in FIG. **6**, the translational movement is output in line with the actuator. FIG. **6** shows an EMA with an in-line translating axle **212**.

[0045] Another option for driving the FCV piston **104** is an electro-hydraulic actuator (EHA) (for example, as shown in the example embodiment of FIG. **7**) coupled with a pump system and a reservoir of fluid. As shown, the EHA includes a piston **280** disposed in a housing **218** such that a first hydraulic chamber **280** is created between one end of the piston **280** and an inner surface of the housing **218** and a second hydraulic chamber **282** is created between the opposite end of the piston **280** and the inner surface of the housing **218**. The piston **280** therefore isolates and seals the hydraulic chambers **282**, **284** from each other. A first hydraulic port **283** extends through the housing **218** to the first chamber **282**, and a second hydraulic port **285** extends through the housing **218** to the second chamber **284**. In use, hydraulic fluid is pumped from the reservoir through the first and/or second port **283**, **285** to the respective chamber **282**, **284**. The piston **280** is connected to the piston **104** of the FCV via the linkage **300**. A piston seal **286** is disposed about the piston **280** proximate to each end of the piston **280**.

[0046] In use, the pump provides pressurized hydraulic fluid to operate the EHA. A manifold can distribute the pressurized hydraulic fluid to one or the other hydraulic chamber **282**, **284** of the actuator. One chamber is used to push the FCV to an open position, the other one to push the FCV to a close position. In other words, flow of hydraulic fluid from the reservoir, through one of the ports **283**, **285** into one of the hydraulic chambers **282**, **284** moves the piston **280** in a direction that thereby moves the piston **104** of the FCV in a direction that opens the FCV, and flow of hydraulic fluid from the reservoir, through the other port **283**, **285** into the other hydraulic chamber **282**, **284** moves the piston **280** in the opposite direction, thereby moving the piston **104** of the FCV in the

opposite direction to close the FCV.

[0047] As shown in FIG. 7, the piston **280** can be equipped with two connecting rods **281**, which are used for the connection to the FCV piston **104**. Alternatively, the connecting rods **281** can be connected to or anchor in the FCV main housing **118** with the hydraulic actuator **200** coupled to the FCV piston **104**. In this implementation, clean hydraulic oil is present on both sides of the hydraulic piston seals **286** to avoid loss of hydraulic fluid (or ingress of well fluids) through leaks around the dynamic seals. A series of bellows **288** isolate the clean hydraulic fluid from the well fluids while permitting movement of the piston **280**. The fluid internal to the bellows **288** is at the same pressure as the annulus, as the bellows **288** may not tolerate a substantial differential pressure. This oil volume is connected to the oil reservoir of the pump system (see hydraulic schematics discussed in greater detail below) through a third port **287**.

[0048] In some configurations, to reduce the number of ports and/or ensure the oil volume internal to the bellows **288** is always connected to the lowest pressure of both hydraulic chambers **282**, **284**, the third port **287** may be replaced by an inverse shuttle valve **290**, as illustrated in FIG. 8. The inverse shuttle valve **290** acts as a logical hydraulic function, putting the exit port (third port **287**) in communication with the lowest pressure port between the chambers **282**, **284**.

[0049] For the configurations illustrated in FIGS. 7 and 8, a pump system **350** equipped with or coupled to a manifold (as shown in FIGS. 9-14 and described herein) is used to supply pressurized hydraulic fluid to one side or the other of the EHA piston (i.e., to the first chamber **282** or the second chamber **284**). The pump system **350** includes a motor and a pump. The manifold includes hydraulic circuitry linking the pump system **350** (e.g., the pump) with the actuator **200**. According to some embodiments, the pump system may rely solely on electric power. Examples include an electric motor coupled to a gear box and a hydraulic pump such as a piston or swashplate pump. The manifold also may include a compensating system **360** (shown in FIGS. 9-14) to equalize the oil reservoir pressure with the annulus pressure. This compensating system can be a piston or a bellow as this can ensure a fully sealed system.

[0050] Referring generally to FIGS. 9-14, three examples of manifolds, or hydraulic circuitry, are presented which use solenoid operated valves (SOVs) and other micro hydraulic components. The first example, illustrated in FIGS. 9-10, comprises a circuit with two 3-way, 2-position normally closed solenoid operated valves. The second example, illustrated in FIGS. 11-12, comprises a circuit with two 2-way, 2-position normally open solenoid operated valves. The third example, illustrated in FIGS. 13-14, comprises a circuit with a single 3-way directional solenoid operated valve.

[0051] In the first example manifold implementation illustrated in FIG. 9, the pump system **350**, including a motor and a pump, provides pressurized fluid from the reservoir **351**. A relief valve **352** protects the hydraulic components from over pressure. Excess pressure cracks the relief valve **352** open and lets fluid return straight to the reservoir. The illustrated configuration includes an optional flow regulator **354**, which can be used to evaluate the displacement of the hydraulic actuator **200** using a time base. The flow regulator **354** outputs a constant flow rate, regardless of the differential pressure across it. This allows for controlling the movement of the EHA by relying on the actuation duration. If the position measurement is realized with a position sensor, the flow regulator **354** is not necessary and can be removed. Two normally closed (as shown in FIG. 9) solenoid operated valves (SOVs) **356a**, **356b** drive the EHA in one or the other direction. A compensation line **358** is represented in dotted line from the EHA to take into account the oil volume protected by the bellow(s) **288** (see third port **287** in FIG. 7).

[0052] FIGS. 10A-10B illustrate four modes of operation for the manifold embodiment of FIG. 9. Specifically, FIG. 10A illustrates actuation of the EHA in an open direction (e.g., moving the EHA piston **280** upwards). The pump system **350** is on or powered up and pumps hydraulic fluid from the reservoir through the manifold. As shown, SOV **356a** is closed, but SOV **356b** is activated to open, so that hydraulic fluid flows through SOV **356b** to the bottom chamber (in the orientation of

FIG. 10A) of the EHA 200, thereby moving the EHA piston 280 upward. As described herein, the actuator 200 is coupled to the FCV piston 104 via a linkage 300, such that movement of the EHA piston 280 thereby causes corresponding movement of the FCV piston 104. FIG. 10B illustrates actuation of the EHA in a close direction (e.g., moving the EHA piston 280 downwards). The pump system 350 is on or powered up, SOV 356b is closed, and SOV 356a is activated to open, so that hydraulic fluid flows through SOV 356a to the top chamber (in the orientation of FIG. 10B) of the EHA 200, thereby moving the EHA piston 280 downward.

[0053] FIGS. 10C and 10D illustrate mechanical intervention modes. As shown, a shifting tool 400 can be used for mechanical intervention. FIG. 10C illustrates mechanical intervention or override to open the FCV (e.g., moving the piston 280 upwards via upward movement of the shifting tool 400). FIG. 10D illustrates mechanical intervention or override to close the FCV (e.g., moving the piston 280 downwards via downward movement of the shifting tool 400). In both mechanical intervention modes, the pump system 350 is off or powered down, and both SOVs 356a, 356b are closed. Mechanical movement of the piston 280 by the shifting tool 400 forces circulation of hydraulic fluid through the SOVs 356a, 356b from one chamber of the EHA to the other.

[0054] An example of an FCV actuation sequence or method includes the steps of: 1. Power up motor of the pump system 350 such that the pump generates pressure in the hydraulic circuitry up to a max of P.sub.r (cracking pressure of the relief valve); 2. Activate the desired SOV 356a, 356b so the EHA 200 starts moving; 3. De-activate the activated SOV to stop the EHA 200 movement; and 4. Stop the motor and pump (or pump system 350). This circuitry is compatible with mechanical intervention as both EHA hydraulic chambers 282, 284 are in direct communication when the SOVs 356a, 356b are not activated, thus allowing EHA piston 280 movement without hydraulic lock.

[0055] In the second example manifold implementation illustrated in FIG. 11, the hydraulic circuitry is a slight variation of the circuitry illustrated in FIG. 9. Instead of 3-way, 2-position, normally closed SOVs (as included in the manifold of FIGS. 9-10), the manifold of FIG. 11 includes 2-way, 2-position, normally open (as shown in FIG. 11) SOVs 366a, 366b, plus the addition of an inverse shuttle valve 290 for releasing the low pressure side of the EHA hydraulic piston 280 to the reservoir and pressure compensator or compensation bellows 360. The circuitry is compatible with mechanical intervention as both sides of the EHA piston 280 are in communication when the SOVs 366a, 366b are not actuated. This embodiment utilizes one additional hydraulic component (inverse shuttle valve 290) but has the advantage of using simpler and potentially more reliable SOVs 366a, 366b.

[0056] FIGS. 12A-12D illustrate four modes of operation for the manifold of FIG. 11. FIG. 12A illustrates actuation of the EHA piston 280 in an open direction (e.g., moving the EHA piston 280 upwards). The pump system 350 is on or powered up and pumps hydraulic fluid from the reservoir through the manifold. As shown, SOV 366b is in its default open position, but SOV 366a is activated to close, so that hydraulic fluid flows through SOV 366b to the bottom chamber (in the orientation of FIG. 12A) of the EHA 200, thereby moving the EHA piston 280 upward. As described herein, the actuator 200 is coupled to the FCV piston 104 via a linkage 300, such that movement of the EHA piston 280 thereby causes corresponding movement of the FCV piston 104. FIG. 12B illustrates actuation of the EHA 200 in a close direction (e.g., moving the EHA piston 280 downwards). The pump system 350 is on or powered up, SOV 366a is in its default open position, and SOV 366b is activated to close, so that hydraulic fluid flows through SOV 366a to the top chamber (in the orientation of FIG. 12B) of the EHA 200, thereby moving the EHA piston 280 downward.

[0057] FIGS. 12C and 12D illustrate mechanical intervention modes. As shown, shifting tool 400 can be used for mechanical intervention. FIG. 12C illustrates mechanical intervention or override to open the FCV (e.g., moving the piston 280 upwards via upward movement of the shifting tool 400). FIG. 12D illustrates mechanical intervention or override to close the FCV (e.g., moving the

piston **280** downwards via downward movement of the shifting tool **400**). In both mechanical intervention modes, the pump system **350** is off or powered down, and both SOVs **366a**, **366b** are open. Mechanical movement of the piston **280** by the shifting tool **400** forces circulation of hydraulic fluid through the SOVs **366a**, **366b** from one chamber of the EHA to the other.

[0058] An example of an FCV actuation sequence or method of the embodiment of FIGS. **11-12** includes the steps of: 1. Activate the desired SOV **366a**, **366b** first. At this stage there is no EHA **200** movement as there is no pressure in the system; 2. Power up motor of the pump system **350** such that the pump generates pressure that starts actuating the EHA **200** and associated FCV piston **104**; 3. Stop the motor and pump such that the EHA **200** stops, as well as the associated FCV **104**; and 4. De-activate the SOV.

[0059] In the third example manifold implementation illustrated in FIG. **13**, hydraulic circuitry is illustrated which uses a single SOV **376** as a directional switch. If the SOV **376** is not energized, the system will move the EHA **200** towards the open position as soon as the pump system **350** is activated. To actuate the EHA **200** in the other (close) direction, the SOV **376** is energized. The implementation illustrated in FIG. **13** can be reversed such that movement of the EHA **200** is to close when the SOV **376** is not activated.

[0060] To be compatible with mechanical intervention, an additional relief valve **372** is used as illustrated in FIGS. **13-14**. To mechanically operate the FCV with a shifting tool **400**, the operator applies an amount of force that will create pressure in the hydraulic system high enough to crack open the relief valves **352**, **372**. The relief valves **352**, **372** and the EHA piston **280** area can be sized such that the effort to operate the valve mechanically is compatible with the different shifting method used (e.g., slickline, or tractor). For reference, the Schlumberger tractor ReSOLVE® can apply up to 40,000 lbfs linearly. This should far exceed the load desired for operating the FCV piston **104** manually.

[0061] FIGS. **14A-14D** illustrate four modes of operation for the manifold of FIG. **13**. FIG. **14A** illustrates actuation of the EHA piston **280** in an open direction (e.g., moving the EHA piston **280** upwards). The pump system **350** is on or powered up and pumps hydraulic fluid from the reservoir through the manifold. As shown, SOV **376** is in its default position so that hydraulic fluid flows through SOV **376** to the bottom chamber (in the orientation of FIG. **14A**) of the EHA **200**, thereby moving the EHA piston **280** upward. As described herein, the actuator **200** is coupled to the FCV piston **104** via a linkage **300**, such that movement of the EHA piston **280** thereby causes corresponding movement of the FCV piston **104**. FIG. **14B** illustrates actuation of the EHA **200** in a close direction (e.g., moving the EHA piston **280** downwards). The pump system **350** is on or powered up, SOV **376** is activated, so that hydraulic fluid flows through SOV **376** to the top chamber (in the orientation of FIG. **14B**) of the EHA **200**, thereby moving the EHA piston **280** downward.

[0062] FIGS. **14C** and **14D** illustrate mechanical intervention modes. As shown, shifting tool **400** can be used for mechanical intervention. FIG. **14C** illustrates mechanical intervention or override to open the FCV (e.g., moving the piston **280** upwards via upward movement of the shifting tool **400**). FIG. **14D** illustrates mechanical intervention or override to close the FCV (e.g., moving the piston **280** downwards via downward movement of the shifting tool **400**). In both mechanical intervention modes, the pump system **350** is off or powered down, and the SOV **376** is in its default state. As described, the operator applies sufficient force to the shifting tool **400** to create pressure in the manifold high enough to open the relief valves **352**, **372** such that hydraulic fluid flows through the circuit from one chamber of the EHA to the other.

[0063] An example of an FCV actuation sequence or method for opening the valve of the embodiment of FIGS. **13-14** includes the steps of: 1. Power up motor of the pump system **350** such that the pump generates pressure that starts actuating the EHA **200** and associated FCV piston **104** towards the open direction; 2. Stop the motor and pump; the EHA **200** stops as well as the associated FCV. An example of an FCV actuation sequence or method for closing the valve

includes the steps of: 1. Activate the SOV **376** first. At this stage, no EHA movement has occurred as there is no pressure in the system; 2. Power up motor of the pump system **350** such that the pump generates pressure that starts actuating the EHA and associated FCV piston towards the closed position; 3. Stop the motor and pump; the EHA stops as well as the associated FCV; and 4. De-activate the SOV **376**.

[0064] With respect to position measurement, the measurement of the displacement of the piston can be done multiple ways. A first method is by direct measurement of the FCV piston **104** position via a position sensor (e.g. LVDT, resistive, AMR, acoustic, or other appropriate sensor). The position sensor, e.g., sensor **240**, can be located in its own groove in the FCV main housing **118** in parallel to the actuator **200** and other electronics **230**, as shown in FIG. **3**.

[0065] Other methods of position measurement also may be employed, such as providing measurement components inside the actuator **200**. Examples include: 1. A resolver counting motor turns in the EMA can provide displacement information of the mechanical actuator. This can translate directly to the FCV piston **104** position once the position measurement is calibrated (record the full close position for instance). 2. Time-based actuation for the electro hydraulic actuator: each of the three illustrated hydraulic circuit embodiments includes a flow regulator **354** that outputs a constant flowrate regardless of the differential pressure across it. With the information of the hydraulic fluid rate flowing to the EHA piston chamber it is straightforward to determine the displacement of the actuator as a function of the actuation duration. Once the system is calibrated, the actual FCV position can be computed easily.

[0066] Depending on the embodiment, various types of linkages **300** may be used between the FCV piston **104** and the electrically powered actuator **200**. For example, with an electro hydraulic actuator **200**, the linkage **300** between the FCV piston **104** and the actuator **200** itself can be a straight anchoring. This will provide a simple technical solution for transmitting the load and displacement from the actuator **200** to the piston **104**.

[0067] As the hydraulic circuitry embodiments described herein are compatible with mechanical intervention, the FCV piston **104** can be operated with a shifting tool **400** while still connected to the actuator **200**. The actuator **200** will not create hydraulic lock which could otherwise prevent the mechanical override of the FCV. The embodiment of hydraulic circuitry shown in FIGS. **13-14** (single SOV **376** design) may utilize extra force to shift the piston due to cracking pressure of the relief valves **352, 372**.

[0068] When the FCV is equipped with an electro mechanical actuator **200**, there may be a desire to unlatch the actuator **200** from the piston **104**. Unlatching permits overriding mechanically the valve position without damaging the actuator **200** in case the drive screw is not reversible (i.e. the assembly of the screw, gearbox, and motor will not rotate back regardless of the load applied on the actuator axles). In this particular case, the linkage mechanism **300** should include a releasable latching system such as a collet or a disengaging system. Examples of two embodiments include: 1. A shear system. A piece in the linkage **300** will break at a controlled load exceeding the nominal operating load of the actuator **200**, thus releasing the piston **104** from the actuator **200**. An example of such shear system is the shear pin used in packers, breaking at a specified effort; and 2. An elastic latch system that will disengage once the axial load exceeds the latching force. The latch can be re-engaged later by moving the piston manually or operating the actuator if its function is not lost.

[0069] Although a few embodiments of the disclosure have been described in detail above, those of ordinary skill in the art will readily appreciate that many modifications are possible without materially departing from the teachings of this disclosure. Accordingly, such modifications are intended to be included within the scope of this disclosure as defined in the claims.

Claims

- 1.** A method of operating a flow control valve, comprising: powering up a pump system configured to pump hydraulic control fluid from a reservoir; activating a selected solenoid operated valve (SOV) in a manifold comprising hydraulic circuitry linking the pump system with an electro-hydraulic actuator mounted externally to the flow control valve; flowing hydraulic control fluid from the reservoir, through the manifold, and into a chamber of the electro-hydraulic actuator such that a piston of the electro-hydraulic actuator moves in an open or a close direction; and moving a piston of the flow control valve by movement of the piston of the electro-hydraulic actuator.
- 2.** The method of claim 1, wherein the SOV is a 2-way, 2-position, normally open valve.
- 3.** The method of claim 1, wherein the electro-hydraulic actuator is held in place along an outer surface of a housing of the flow control valve with one or more clamps or protectors.
- 4.** The method of claim 1, wherein the SOV acts as a directional switch.
- 5.** The method of claim 1, further comprising performing mechanical intervention on the electro-hydraulic actuator by using a shifting tool to mechanically move the piston of the electro-hydraulic actuator.
- 6.** The method of claim 1, wherein the flow control valve comprises a housing, wherein the piston is moveably disposed within the housing; and the electro-hydraulic actuator is connected to the piston via a linkage, the electro-hydraulic actuator responding to electrical inputs to shift the internal piston to flow positions.
- 7.** The method of claim 6, wherein the electro-hydraulic actuator is held in place along an outer surface of the housing of the flow control valve with one or more clamps or protectors.
- 8.** The method of claim 6, wherein the electro-hydraulic actuator is disposed in a groove formed in an outer surface of the housing of the flow control valve.
- 9.** The method of claim 6, wherein an outer surface of the housing comprises one or more grooves formed therein.
- 10.** The method of claim 9, wherein the one or more grooves comprises a first groove housing the electro-hydraulic actuator and a second groove housing electronics or one or more sensors.
- 11.** The method of claim 6, wherein the electro-hydraulic actuator comprises an electro-mechanical actuator (EMA).
- 12.** The method of claim 6, wherein the electro-hydraulic actuator comprises an electro-hydraulic actuator (EHA).
- 13.** The method of claim 12, further comprising a manifold and a pump system comprising a motor and a pump, the manifold comprising hydraulic circuitry linking the pump system to the electro-hydraulic actuator, and the pump system configured to pump hydraulic control fluid from a reservoir through the manifold to the electro-hydraulic actuator.
- 14.** The method of claim 13, the manifold comprising at least one solenoid operated valve (SOV).
- 15.** The method of claim 12, wherein mechanical intervention for mechanically shifting the flow control valve is configured to be performed while the electro-hydraulic actuator is connected to the internal piston.
- 16.** The method of claim 6, wherein the flow control valve is mounted along a well tubing, the flow control valve having a flow area equivalent to an internal cross-sectional area of the well tubing.
- 17.** The method of claim 6, wherein the linkage is configured to be disconnected to enable mechanical intervention for mechanically shifting the flow control valve.
- 18.** A flow control valve comprising: a housing; a piston movably disposed within the housing to adjust flow through the flow control valve; at least one groove formed in an outer surface of the housing, wherein the at least one groove comprises a first groove housing an electro-hydraulic actuator and a second groove housing electronics; and a solenoid operated valve (SOV) in a manifold, the manifold comprising hydraulic circuitry and a linkage coupling the electro-hydraulic actuator to the piston such that movement of the electro-hydraulic actuator causes movement of the piston.

- 19.** The flow control valve of claim 18, wherein the electro-hydraulic actuator is held in place along the outer surface of the housing of the flow control valve with one or more clamps or protectors.
- 20.** The flow control valve of claim 18, wherein the electro-hydraulic actuator comprises an electro-hydraulic actuator comprising an internal piston, wherein movement of the internal piston of the electro-hydraulic actuator causes movement of the piston of the flow control valve to adjust flow through the flow control valve.
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